

Nerve Signals

The brain and nervous system work by sending signals through cells as pulses of electrical charge and between cells either by using chemical messengers called neurotransmitters or by electric charge.

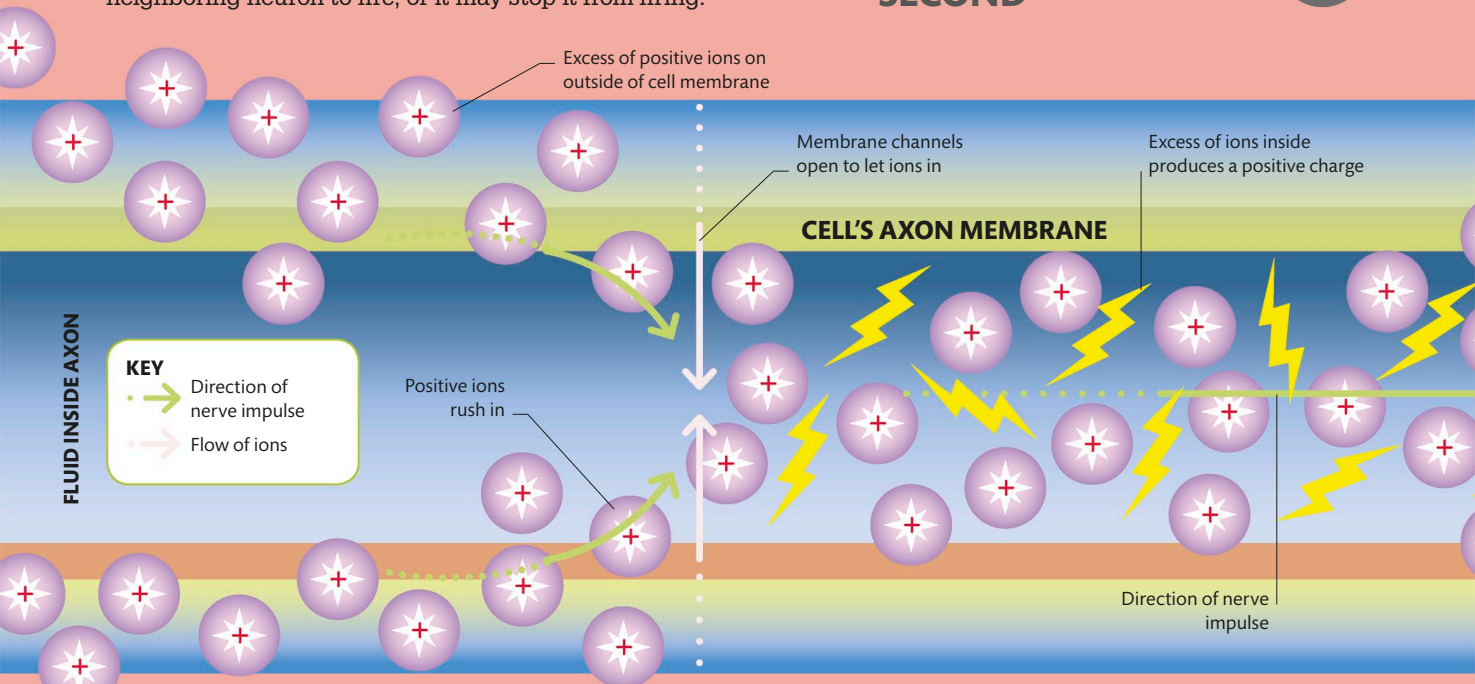
Action potential

Neurons signal by creating an action potential—a surge of electricity created by sodium and potassium ions crossing the cell's membrane. It travels down the axon and stimulates receptors on dendrites of neighboring cells. The junction between cells is called a synapse. In many neurons, the charge is carried over a minute gap between axon and dendrite by chemicals, called neurotransmitters, released from the tip of the axon. These junctions are known as chemical synapses. The signal may cause the neighboring neuron to fire, or it may stop it from firing.

HOW DOES A NERVE COMMUNICATE DIFFERENT INFORMATION?

Receiving cells have different types of receptors, which respond to different neurotransmitters. The “message” differs according to which neurotransmitters are sent and received and in what quantities.

SOME NERVE IMPULSES TRAVEL FASTER THAN 330 FT (100 M) PER SECOND



1

Resting potential

When the neuron is at rest, there are more positive ions outside the membrane than inside. This causes a difference in polarization, or electrical potential, across the membrane called the resting potential. The difference is about -70 millivolts, meaning the outside is positive.

2

Depolarization

Chemical changes from the cell body allow positive ions to flood into the cell through the membrane. That reverses the polarization of the axon, making the potential difference +30 millivolts.